

Alexander C. Flinchum

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EXPERIENCE

GENERAL MOTORS

Austin, TX

Software Engineer II

Feb 2024 – May 2026

- Led automated testing integration within GM's DaaS Java/Spring Boot platform, increasing automated regression coverage from 35% to 90% and reducing manual QA validation effort by 40% across release cycles
- Built and maintained backend services and IMOS API integrations in Java, .NET, React, and GraphQL, leveraging Copilot to support 500+ plant users across 100+ facilities managing equipment metadata and node relationships
- Developed reusable GraphQL query and service integration patterns adopted across 3+ manufacturing business units, improving backend data retrieval consistency and accelerating API onboarding for new teams by 25%
- Supported production stability through RedHat OpenShift migration by implementing smoke testing and post-deployment validation workflows across 10+ services, maintaining 99%+ reliability throughout transition

Software Engineer I, Test Automation

Aug 2020 – Feb 2024

- Built reusable Java and C# test automation frameworks and API validation workflows from scratch, reducing manual regression testing by 60% and cutting defect escape rate by 25% across 8+ bi-weekly release cycles annually
- Owned end-to-end release readiness coordination across a 5+ cross-functional team, resolving 15+ production and integration defects per sprint and maintaining enterprise application stability
- Expanded automated test coverage from 30% to 85% across enterprise backend integrations and application workflows, accelerating QA cycle time by 2 days per sprint and enabling more reliable bi-weekly deployments

Software Developer

Jul 2019 – Aug 2020

- Owned and maintained InDepth Lite, an Angular-based file-processing application supporting 150 users across GM's analytics platform, collaborating with frontend, backend, and analytics teams to sustain 99%+ reliability
- Modernized 100% of legacy Angular components during a 3-month framework migration, resolving deprecated dependencies, compatibility issues, and frontend regressions with zero disruption to active production workflows
- Identified and resolved 15+ frontend UI defects, reducing user-reported support tickets by 30% and improving application stability across active development cycles

EDUCATION

BOSTON UNIVERSITY

Boston, MA

M.S. in Software Development

May 2025

- Coursework:** Data Structures and Algorithms, Web Application Development, Software Design and Patterns, Advanced Programming Techniques, Database Design and Implementation for Business, Software Engineering

INDIANA UNIVERSITY

Bloomington, IN

B.S. in Informatics (Distinction); Minor in Computer Science, Cognate in Game Design; (GPA 3.8/4.0)

May 2019

- Honors:** National Society of Collegiate Scholars, Hutton Honors College Inductee
- Awards:** Dean's List Honors, 3-time Recipient of Indiana University Founders Scholar

TECHNICAL SKILLS & PROJECTS

Languages: Java, Python 3, C#, .NET, GraphQL | **Web Development:** JavaScript, HTML, CSS, Angular, ReactJS

Databases: SQL, Postgres | **Other:** GitHub, Jira, Browserstack, Test Automation, CI/CD, API, OOP

University Course Management System

- Designed and implemented a modular Java-based course enrollment system using object-oriented design patterns, scalable class hierarchies, reusable interfaces, and UML architecture modeling for academic management workflows
- Built course registration and notification features using the Observer pattern to automate enrollment workflows across student, faculty, and department systems, leveraging Gemini to accelerate UML planning and iteration

Video Game Collection Database System

- Designed and implemented a normalized relational database system using SQL Server, featuring indexed foreign keys, ERD-based schema modeling, stored procedures, and triggers to support tracking and reporting workflow

ADDITIONAL INFORMATION

Community: Austin Symphony Orchestra (Usher), Austin Humane Society (Animal Socializer), Texas Book Festival

Interests: Running (Run Club Social Member), Volleyball, IU Sports, Silent Book Club, Video Game Design